DSC 190/291 Assignment 2

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Contents

Part A: Restriction Patterns for Two Intervals

Problem Overview

In Part A we study the hypothesis class consisting of unions of at most two intervals on the real line. Let

$$\mathcal{H}_{2\text{int}} = \{\mathbf{1}_{I_1 \cup I_2} : I_1, I_2 \text{ intervals in } \mathbb{R}\},$$

where either interval is allowed to be empty. When we restrict a hypothesis in $\mathcal{H}_{2\text{int}}$ to an ordered sample

$$x_1 < x_2 < \cdots < x_n,$$

we obtain a binary string

$$\big(h(x_1), \ldots, h(x_n)\big) \in \{0,1\}^n.$$

The main goal of Part A is to understand exactly which binary strings can arise, count them, determine the VC dimension, and compare the exact counts to general bounds from learning theory.

Problem A1: Restriction Patterns

Question.
Characterize exactly which binary labelings on an ordered sample

$$x_1 < x_2 < \cdots < x_n$$

are realizable by a hypothesis in $\mathcal{H}_{2\text{int}}$.

The key statement to prove is:

$$\text{a labeling is realizable} \quad \Longleftrightarrow \quad \text{it has at most two runs of 1s.}$$

Restatement

We are given n ordered points on the real line and a binary labeling

$$y = (y_1, \ldots, y_n) \in \{0,1\}^n.$$

We must determine exactly when there exists a union of at most two intervals whose indicator function agrees with this labeling on the sample.

Explanation

An interval intersects an ordered sample in a contiguous block of indices. Since our class allows at most two intervals, the positive labels should form at most two contiguous blocks. In binary-string language, contiguous blocks of positive labels are exactly runs of 1s. So the whole problem is to make precise the statement that unions of two intervals correspond exactly to binary strings with at most two runs of 1s.

Solution Plan

1. Prove the forward implication: every hypothesis in $\mathcal{H}_{2\text{int}}$ induces at most two runs of 1s.
2. Prove the reverse implication: every binary string with at most two runs of 1s can be realized by explicitly constructing suitable intervals.
3. Conclude the equivalence.

Formal Solution / Proof

Theorem 1. *Let $x_1 < \cdots < x_n$ be distinct real numbers. A labeling*

$$(y_1, \ldots, y_n) \in \{0,1\}^n$$

is realizable by $\mathcal{H}_{2\text{int}}$ if and only if the binary string

$$y_1 y_2 \cdots y_n$$

contains at most two runs of 1*s.*

Proof.
($\Rightarrow$) Suppose $h \in \mathcal{H}_{2\text{int}}$.

Then there exist intervals I_1 and I_2 such that

$$h = \mathbf{1}_{I_1 \cup I_2}.$$

For a sorted sample, the set of indices i with $x_i \in I_1$ is either empty or a contiguous block. Indeed, once the sample enters an interval, it cannot leave and later re-enter that same interval while the sample points are ordered increasingly. The same statement is true for I_2.

Therefore the set of indices with label 1 under h is the union of at most two contiguous blocks. Equivalently, the binary string

$$h(x_1) \cdots h(x_n)$$

has at most two runs of 1s.

($\Leftarrow$) Suppose the string $y_1 \cdots y_n$ has at most two runs of 1s.

We construct a hypothesis in $\mathcal{H}_{2\text{int}}$ that realizes it.

Case 1: no runs of 1s.

Then every label is 0. This is realized by taking the union of two empty intervals.

Case 2: exactly one run of 1s.

Suppose the run occupies indices $a, \ldots, b$. Then define one interval I_1 that contains exactly the sample points $x_a, \ldots, x_b$. For example, if $1 < a$ and $b < n$, we may take

$$I_1 = \left[\frac{x_{a-1} + x_a}{2}, \frac{x_b + x_{b+1}}{2} \right].$$

If the run touches the left or right edge of the sample, we modify the corresponding endpoint in the obvious way. Let I_2 be empty. Then

$$\mathbf{1}_{I_1 \cup I_2}(x_i) = y_i \qquad \text{for all } i.$$

Case 3: exactly two runs of 1s.

Suppose the two runs occupy blocks $a, \ldots, b$ and $c, \ldots, d$, where

$$1 \le a \le b < c \le d \le n.$$

Choose one interval I_1 containing exactly the first block and another interval I_2 containing exactly the second block. Then

$$\mathbf{1}_{I_1 \cup I_2}(x_i) = y_i \qquad \text{for all } i.$$

Thus every binary string with at most two runs of 1s is realizable by a union of at most two intervals. $\square$

Conclusion

The realizable restriction patterns of $\mathcal{H}_{2\text{int}}$ are exactly the binary strings with at most two runs of 1s.

Lean Formalization

Lean: NO.

Computational Check

For $n \leq 6$, the script `hw2/code/enumeration_check.py` checks that the counting formula from Problem A2 matches the number of binary strings with at most two runs of 1s.

Problem A3: VC Dimension and Halving Bound

Question.
Determine the VC dimension of $\mathcal{H}_{2\text{int}}$. Show that $n = 4$ is shatterable and that $n = 5$ is not. Then state the exact Halving bound and compare it to the Sauer–Shelah bound.

Restatement

We want the largest sample size on which $\mathcal{H}_{2\text{int}}$ can realize every binary labeling. Then we use the exact growth function from Problem A2 to state the Halving mistake bound.

Explanation

The run characterization from Problem A1 makes the VC-dimension argument very clean. Every binary string on four bits has at most two runs of 1s, but the five-bit string 10101 has three runs of 1s and therefore is not realisable.

Solution Plan

1. Show that every labeling of four ordered points is realizable.
2. Exhibit a nonrealizable labeling on five ordered points.
3. Conclude $\text{VCdim}(\mathcal{H}_{2\text{int}}) = 4$.
4. Substitute the exact growth function into $\log_2 \Gamma_{\mathcal{H}_{2\text{int}}}(n)$.

Formal Solution / Proof

Claim 2.

$$\text{VCdim}(\mathcal{H}_{2\text{int}}) = 4.$$

Proof.
Take any four ordered points

$$x_1 < x_2 < x_3 < x_4.$$

Any binary string of length 4 has at most two runs of 1s. Therefore by Problem A1 every labeling on these four points is realizable by $\mathcal{H}_{2\text{int}}$. So $\{x_1, x_2, x_3, x_4\}$ is shattered and

$$\text{VCdim}(\mathcal{H}_{2\text{int}}) \geq 4.$$

Now take any five ordered points

$$x_1 < x_2 < x_3 < x_4 < x_5.$$

The labeling

$$(1, 0, 1, 0, 1)$$

has three runs of 1s. By Problem A1 this labeling is not realizable by a union of at most two intervals. Therefore no 5-point set is shattered, so

$$\mathrm{VCdim}(\mathcal{H}_{2\mathrm{int}}) \le 4.$$

Hence

$$\mathrm{VCdim}(\mathcal{H}_{2\mathrm{int}}) = 4.$$

□

From Problem A2,

$$\Gamma_{\mathcal{H}_{2\mathrm{int}}}(n) = 1 + \binom{n+1}{2} + \binom{n+1}{4}.$$

Therefore the exact Halving bound is

$$M_{\mathrm{Halving}} \le \log_2\left(1 + \binom{n+1}{2} + \binom{n+1}{4}\right).$$

Since $\mathrm{VCdim}(\mathcal{H}_{2\mathrm{int}}) = 4$, Sauer–Shelah yields

$$\Gamma_{\mathcal{H}_{2\mathrm{int}}}(n) \le \sum_{k=0}^{4} \binom{n}{k},$$

so the generic VC-based Halving bound is

$$M_{\mathrm{Halving}} \le \log_2\left(\sum_{k=0}^{4} \binom{n}{k}\right).$$

Conclusion

The VC dimension is exactly

$$\mathrm{VCdim}(\mathcal{H}_{2\mathrm{int}}) = 4,$$

and the exact Halving bound is sharper than the generic Sauer–Shelah-based bound because it uses the full structure of the class.

Lean Formalization

Lean: NO.

Computational Check

The exact growth values satisfy

$$\Gamma_{\mathcal{H}_{2\mathrm{int}}}(4) = 16 = 2^4, \qquad \Gamma_{\mathcal{H}_{2\mathrm{int}}}(5) = 31 < 32 = 2^5,$$

which agrees with the VC-dimension conclusion.

Problem A4: Tightness of Sauer–Shelah

Question.
Construct a hypothesis class with VC dimension exactly d and growth function

$$\sum_{k=0}^{d} \binom{n}{k}.$$

Explain why this shows the Sauer–Shelah bound can be tight.

Restatement

We need an explicit class whose growth function exactly equals the Sauer–Shelah upper bound.

Explanation

The natural example is the class of indicator functions of subsets of size at most d. On an n-point sample, the realizable labelings are exactly those with at most d ones.

Solution Plan

1. Define $\mathcal{H}_{\leq d}$.
2. Show that every d-point set is shattered.
3. Show that no $(d+1)$-point set is shattered.
4. Count the realizable labelings on an n-point sample.

Formal Solution / Proof

Let X be an infinite set and define

$$\mathcal{H}_{\leq d} = \{\mathbf{1}_S : S \subseteq X,\ |S| \leq d\}.$$

Claim 3.

$$\mathrm{VCdim}(\mathcal{H}_{\leq d}) = d.$$

Proof.
Take any d distinct points $x_1, \ldots, x_d \in X$. For any labeling $(y_1, \ldots, y_d) \in \{0,1\}^d$, let

$$S = \{x_i : y_i = 1\}.$$

Then $|S| \leq d$, so $\mathbf{1}_S \in \mathcal{H}_{\leq d}$, and it realizes the labeling. Hence every d-point set is shattered, so

$$\mathrm{VCdim}(\mathcal{H}_{\leq d}) \geq d.$$

Now consider $d+1$ distinct points $x_1, \ldots, x_{d+1}$. The all-ones labeling on these points would require a set S containing all $d+1$ points, which is impossible because hypotheses in $\mathcal{H}_{\leq d}$ have support size at most d. Therefore no $(d+1)$-point set is shattered, so

$$\mathrm{VCdim}(\mathcal{H}_{\leq d}) \leq d.$$

Hence

$$\mathrm{VCdim}(\mathcal{H}_{\leq d}) = d.$$

□

On an n-point sample, a labeling is realizable exactly when it has at most d ones. For each k, there are $\binom{n}{k}$ labelings with exactly k ones, so

$$\Gamma_{\mathcal{H}_{\leq d}}(n) = \sum_{k=0}^{d} \binom{n}{k}.$$

Conclusion

This class has VC dimension exactly d and achieves the Sauer–Shelah upper bound exactly. Therefore the inequality can be tight.

Lean Formalization

Lean: NO.

Computational Check

Not needed beyond the direct combinatorial count.

Problem A5: AI Proof Audit

Question.
An AI proof counts possible switch points as if a union of two intervals could realize any binary string with up to four changes. Explain why this is invalid and give the correct statement.

Restatement

We must identify the flaw in a switch-point counting argument and replace it with the correct run-based reasoning.

Explanation

The error is that switch points are not free parameters. The geometry of two intervals forces the positive set to be the union of at most two contiguous blocks, which is a stronger condition than merely bounding the number of label changes.

Solution Plan

1. Explain why arbitrary change-point counting overcounts.
2. State the correct two-run characterization.
3. Deduce the correct growth-function formula.

Formal Solution / Proof

The AI proof is invalid because it ignores interval structure. A union of two intervals does not produce an arbitrary collection of sign changes; it produces at most two contiguous positive blocks. Therefore the correct structural statement is:

$$\text{a labeling is realizable by } \mathcal{H}_{2\text{int}} \iff \text{it has at most two runs of 1s.}$$

Once this is proved, the exact growth function follows from Problem A2:

$$\Gamma_{\mathcal{H}_{2\text{int}}}(n) = 1 + \binom{n+1}{2} + \binom{n+1}{4}.$$

Conclusion

The AI argument overcounts because it counts unconstrained change patterns. The correct argument is based on runs of 1s.

Lean Formalization

Lean: NO.

Computational Check

The small-n enumeration supports the corrected formula.

Part B: Quadratic Thresholds

Problem Overview

In Part B we study

$$\mathcal{H}_{\text{quad}} = \left\{x \mapsto \mathbf{1}[ax^2 + bx + c \geq 0] : (a, b, c) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}\right\}.$$

The main tasks are to represent this class linearly, describe its restriction patterns, count them, and compute its VC dimension.

Problem B1: General Upper Bound for Linear Representations

Question.
Prove that if a hypothesis class is represented as homogeneous linear threshold functions in $\mathbb{R}^D$, then its VC dimension is at most D.

Restatement

Suppose

$$\mathcal{H}_\phi = \left\{x \mapsto \mathbf{1}[\langle w, \phi(x)\rangle \geq 0] : w \in \mathbb{R}^D\right\}.$$

We must prove

$$\text{VCdim}(\mathcal{H}_\phi) \leq D.$$

Explanation

The proof uses linear dependence. If $D+1$ points were shattered, then their feature vectors would realize every binary labeling by a homogeneous hyperplane. But $D + 1$ vectors in $\mathbb{R}^D$ are linearly dependent, and that dependence produces a labeling that no such hyperplane can realize.

Solution Plan

1. Assume $D + 1$ points are shattered.
2. Write a nontrivial dependence relation among their feature vectors.
3. Convert coefficient signs into a target labeling.
4. Derive a contradiction by taking an inner product with the separating weight vector.

Formal Solution / Proof

Theorem 4. *Let X be any domain and $\phi : X \to \mathbb{R}^D$. Define*

$$\mathcal{H}_\phi = \left\{x \mapsto \mathbf{1}[\langle w, \phi(x)\rangle \geq 0] : w \in \mathbb{R}^D\right\}.$$

Then

$$\text{VCdim}(\mathcal{H}_\phi) \leq D.$$

Proof.
Assume for contradiction that $\mathcal{H}_\phi$ shatters

$$x_1, \ldots, x_{D+1} \in X.$$

Let

$$v_i = \phi(x_i) \in \mathbb{R}^D.$$

Since there are $D+1$ vectors in $\mathbb{R}^D$, they are linearly dependent. So there exist scalars $\alpha_1, \ldots, \alpha_{D+1}$, not all zero, such that

$$\sum_{i=1}^{D+1} \alpha_i v_i = 0.$$

Remove any zero coefficients. On the remaining indices define the labeling

$$y_i = \begin{cases} 1, & \alpha_i > 0, \\ 0, & \alpha_i < 0. \end{cases}$$

Because the sample is shattered, there exists $w \in \mathbb{R}^D$ such that

$$\alpha_i > 0 \implies \langle w, v_i \rangle > 0, \qquad \alpha_i < 0 \implies \langle w, v_i \rangle < 0.$$

Now take inner products with w:

$$0 = \left\langle w, \sum_{i=1}^{D+1} \alpha_i v_i \right\rangle = \sum_{i=1}^{D+1} \alpha_i \langle w, v_i \rangle.$$

But every term in this final sum is strictly positive, because α_i and $\langle w, v_i \rangle$ have the same sign. Therefore the sum is strictly positive, which is impossible. This contradiction proves

$$\mathrm{VCdim}(\mathcal{H}_\phi) \leq D.$$

□

Conclusion

Any class represented by homogeneous linear thresholds in $\mathbb{R}^D$ has VC dimension at most D.

Lean Formalization

Lean: YES.

Target theorem:

$$\mathcal{H}_\phi = \{x \mapsto \mathbf{1}[\langle w, \phi(x) \rangle \geq 0] : w \in \mathbb{R}^D\} \implies \mathrm{VCdim}(\mathcal{H}_\phi) \leq D.$$

Repository file: `hw2/lean/vc_dimension_bound.lean`.

Computational Check

Not applicable.

Problem B2: Quadratic Transformation

Question.
Define

$$\phi(x) = (x^2, x, 1).$$

Show that quadratic thresholds become linear thresholds under this map and conclude that $\mathrm{VCdim}(\mathcal{H}_{\mathrm{quad}}) \leq 3$.

Restatement

We must show that

$$ax^2 + bx + c = \langle (a, b, c), (x^2, x, 1) \rangle$$

and then apply Problem B1 with $D = 3$.

Explanation

This is the standard feature-lifting idea: the original class is nonlinear in the scalar input x, but linear in the lifted coordinates $(x^2, x, 1)$.

Solution Plan

1. Write the quadratic polynomial as a dot product in $\mathbb{R}^3$.
2. Observe that the threshold class is now a linear-threshold class.
3. Apply the general VC-dimension upper bound from Problem B1.

Formal Solution / Proof

Define

$$\phi : \mathbb{R} \to \mathbb{R}^3, \qquad \phi(x) = (x^2, x, 1).$$

For $a, b, c \in \mathbb{R}$, let $w = (a, b, c) \in \mathbb{R}^3$. Then

$$\langle w, \phi(x) \rangle = \langle (a, b, c), (x^2, x, 1) \rangle = ax^2 + bx + c.$$

Therefore

$$\mathbf{1}[ax^2 + bx + c \geq 0] = \mathbf{1}[\langle w, \phi(x) \rangle \geq 0].$$

So $\mathcal{H}_{\mathrm{quad}}$ is a homogeneous linear-threshold class in $\mathbb{R}^3$. By Problem B1,

$$\mathrm{VCdim}(\mathcal{H}_{\mathrm{quad}}) \leq 3.$$

Conclusion

The quadratic-threshold class lifts to a linear-threshold class in three dimensions, so

$$\mathrm{VCdim}(\mathcal{H}_{\mathrm{quad}}) \leq 3.$$

Lean Formalization

Lean: YES.

Target identity:

$$ax^2 + bx + c = \langle (a, b, c), (x^2, x, 1) \rangle.$$

Repository file: `hw2/lean/quadratic_embedding.lean`.

Computational Check

Not needed for the algebraic identity itself.

Problem B3: Restriction Patterns

Question.
Describe all binary labelings on an ordered sample

$$x_1 < x_2 < \cdots < x_n$$

that can be realized by thresholding a quadratic polynomial.

Restatement

We must characterize all strings of the form

$$\big(\mathbf{1}[q(x_1) \geq 0], \ldots, \mathbf{1}[q(x_n) \geq 0]\big),$$

where $q(x) = ax^2 + bx + c$.

Explanation

The sign of a quadratic can only change at real roots, and a quadratic has at most two real roots. Therefore the nonnegative set is either all of $\mathbb{R}$, empty, one interval, or the complement of one interval. On a sorted sample these geometric possibilities become binary-pattern constraints.

Solution Plan

1. Analyze sign patterns according to the number of roots and the sign of the leading coefficient.
2. Translate each geometric case to a binary pattern on the sample.
3. Prove the converse by constructing quadratics for each valid pattern.

Formal Solution / Proof

Theorem 5. *On an ordered sample $x_1 < \cdots < x_n$, the realizable 1-sets of $\mathcal{H}_{\text{quad}}$ are exactly:*

1. *the empty set,*
2. *one contiguous block,*
3. *the union of a prefix and a suffix, separated by a nonempty middle block of zeros.*

Proof.
Let $q(x) = ax^2 + bx + c$.

If $a = 0$, then q is linear or constant, so the set $\{x : q(x) \geq 0\}$ is either empty, all of $\mathbb{R}$, a left half-line, or a right half-line. On an ordered sample this yields all zeros, all ones, a prefix of ones, or a suffix of ones.

Now suppose $a \neq 0$. A quadratic has at most two real roots.

If there are no real roots, the sign never changes, so the labeling is either all zeros or all ones.

If there are two distinct real roots $r_1 < r_2$, then:

- if $a < 0$, then $q(x) \geq 0$ exactly on $[r_1, r_2]$, which gives one contiguous block of ones on the sample;
- if $a > 0$, then $q(x) \geq 0$ exactly outside (r_1, r_2), which gives a prefix together with a suffix.

Thus every realizable pattern has one of the three stated forms.

Conversely, the empty labeling is realized by $q(x) = -1$, and the all-ones labeling by $q(x) = 1$.

If we want one contiguous block from x_a through x_b, choose roots

$$r_1 < x_a < x_b < r_2$$

with r_1 and r_2 placed in the neighboring gaps, and define

$$q(x) = -(x - r_1)(x - r_2).$$

Then $q(x) \geq 0$ exactly on the chosen block.

If we want a prefix together with a suffix, choose an interior zero block from x_a through x_b, place roots around it, and define

$$q(x) = (x - r_1)(x - r_2).$$

Then $q(x) \geq 0$ exactly outside the middle interval, so the sample labeling is the desired prefix–suffix pattern. □

Conclusion

Quadratic thresholds realize exactly three kinds of positive sets on an ordered sample: empty, one block, or prefix-plus-suffix.

Lean Formalization

Lean: NO.

Computational Check

The small-n enumeration checks that counting these three families produces the exact growth formula in Problem B4.

Problem B4: Growth Function and VC Dimension

Question.
Using the restriction-pattern description from Problem B3, compute the exact growth function of $\mathcal{H}_{\text{quad}}$, determine its VC dimension, and compare the exact count with the Sauer–Shelah bound.

Restatement

We must count exactly how many labelings arise from the three valid pattern types in Problem B3 and then use the result to determine the VC dimension.

Explanation

The counting is clean once the structure is correct. The main subtlety is that a prefix–suffix labeling is determined by its middle zero block, which must be a nonempty contiguous interior block.

Solution Plan

1. Count the empty labeling.
2. Count one contiguous block of ones.
3. Count prefix–suffix patterns.
4. Add the counts and simplify.
5. Check the formula at $n = 3$ and $n = 4$ to get the VC dimension.

Formal Solution / Proof

By Problem B3, realizable labelings fall into three disjoint classes.

Case 1: the empty labeling.

There is exactly one such labeling.

Case 2: one contiguous block of 1s.

Such a labeling is specified by choosing indices

$$1 \le a \le b \le n$$

and labeling exactly $x_a, \dots, x_b$ by 1. Therefore the number of such labelings is

$$\binom{n+1}{2}.$$

Case 3: a prefix together with a suffix.

Such a labeling is determined by a nonempty contiguous middle block of zeros lying strictly inside the sample. If the zero block runs from index a to index b, then

$$2 \le a \le b \le n-1.$$

The number of such contiguous blocks in $\{2, \dots, n-1\}$ is

$$\binom{n-1}{2}.$$

Therefore

$$\Gamma_{\mathcal{H}_{\text{quad}}}(n) = 1 + \binom{n+1}{2} + \binom{n-1}{2}.$$

Simplifying,

$$\begin{aligned}
\Gamma_{\mathcal{H}_{\text{quad}}}(n) &= 1 + \frac{n(n+1)}{2} + \frac{(n-1)(n-2)}{2} \\
&= 1 + \frac{n^2 + n + n^2 - 3n + 2}{2} \\
&= 1 + \frac{2n^2 - 2n + 2}{2} \\
&= n^2 - n + 2.
\end{aligned}$$

Now compute the VC dimension. At $n = 3$,

$$\Gamma_{\mathcal{H}_{\text{quad}}}(3) = 3^2 - 3 + 2 = 8 = 2^3,$$

so three points are shattered. At $n = 4$,

$$\Gamma_{\mathcal{H}_{\text{quad}}}(4) = 4^2 - 4 + 2 = 14 < 16 = 2^4,$$

so four points are not shattered. Hence

$$\text{VCdim}(\mathcal{H}_{\text{quad}}) = 3.$$

Since $\mathrm{VCdim}(\mathcal{H}_{\mathrm{quad}}) = 3$, Sauer–Shelah gives

$$\Gamma_{\mathcal{H}_{\mathrm{quad}}}(n) \leq \sum_{k=0}^{3} \binom{n}{k}.$$

The exact count $n^2 - n + 2$ is sharper because it uses the actual structure of quadratic thresholds.

Conclusion

The exact growth function is

$$\Gamma_{\mathcal{H}_{\mathrm{quad}}}(n) = n^2 - n + 2,$$

and the VC dimension is exactly

$$\mathrm{VCdim}(\mathcal{H}_{\mathrm{quad}}) = 3.$$

Lean Formalization

Lean: OPTIONAL.

Computational Check

The script verifies the values

$$2,\ 4,\ 8,\ 14,\ 22,\ 32$$

for $n = 1, \ldots, 6$.

Problem B5: AI Proof Audit

Question.
An AI proof counts arbitrary change points and concludes a growth function for quadratic thresholds. Explain the flaw and replace it with the correct root-based reasoning.

Restatement

We must diagnose why counting change points alone is mathematically insufficient for quadratic thresholds.

Explanation

The issue is that not every pattern with a small number of sign changes comes from a quadratic. The shape of the polynomial matters: the positive set must be an interval or the complement of an interval.

Solution Plan

1. Explain why free change-point counting overcounts.
2. State the actual structural constraint from Problem B3.
3. Deduce the correct growth function from Problem B4.

Formal Solution / Proof

The AI proof is invalid because it counts change locations as if they could be chosen independently. That ignores the geometry of quadratics. A quadratic polynomial has at most two real roots, and depending on the sign of the leading coefficient, the region where $q(x) \geq 0$ is either one interval or the complement of one interval.

Therefore the correct restriction patterns are not "all strings with at most two changes." The correct statement from Problem B3 is that the realizable 1-sets are exactly:

- the empty set,
- one contiguous block,
- a prefix together with a suffix.

Using this classification, the correct growth function is

$$\Gamma_{\mathcal{H}_{\text{quad}}}(n) = n^2 - n + 2.$$

Conclusion

The flawed AI proof ignores the shape constraints imposed by quadratics. The correct proof must be root-based rather than arbitrary-change-based.

Lean Formalization

Lean: NO.

Computational Check

The enumeration script supports the corrected formula on small values of n.

Part C: AI Usage Report

Question.
Write a short report describing where AI was used, an accepted suggestion, a rejected or modified suggestion, how correctness was verified, and how the workflow improved.

Restatement

This part asks for a reflective description of how AI contributed to the work without replacing mathematical verification.

Explanation

The goal is to show judgment: where AI was helpful, where it needed correction, and what independent checks were used to make sure the final results were correct.

Solution Plan

1. Describe where AI was used in the workflow.
2. Give one accepted suggestion.
3. Give one rejected or revised suggestion.
4. Explain the verification process.
5. Summarize workflow improvements.

Formal Solution / Proof

For this assignment, I used AI mainly for organization, proof drafting, and small computational support. In the proof-writing stage, AI helped turn rough ideas into a cleaner structure with definitions, claims,

and proofs. In the verification stage, AI helped suggest a small brute-force enumeration script for checking the growth-function formulas on small values of n. I also used AI to identify which parts of the assignment were the most natural targets for Lean formalization.

One suggestion I accepted was the idea of using a small-n computational check for the exact growth formulas. I accepted this because the formulas in Parts A and B are exact, so enumeration on $n \leq 6$ is a meaningful sanity check. It does not replace proof, but it helps catch overcounting and missed cases.

One suggestion I rejected or heavily modified was an earlier proof sketch that counted label changes too loosely. That approach risked overcounting because it did not properly encode interval structure in Part A or root structure in Part B. I replaced it with arguments based on runs of 1s for the two-interval class and on the sign pattern of quadratics for the quadratic-threshold class.

I verified correctness in several ways. First, every counting formula was derived from a structural characterization proved immediately before it. Second, I checked the formulas computationally on small samples:

$$\Gamma_{\mathcal{H}_{2\text{int}}}(n) = 1 + \binom{n+1}{2} + \binom{n+1}{4}, \qquad \Gamma_{\mathcal{H}_{\text{quad}}}(n) = n^2 - n + 2.$$

Third, I cross-checked the VC-dimension conclusions against the critical growth values:

$$\Gamma_{\mathcal{H}_{2\text{int}}}(4) = 16,\ \Gamma_{\mathcal{H}_{2\text{int}}}(5) = 31, \qquad \Gamma_{\mathcal{H}_{\text{quad}}}(3) = 8,\ \Gamma_{\mathcal{H}_{\text{quad}}}(4) = 14.$$

Finally, for Lean, I focused on the mathematically central statements that are most reusable rather than trying to formalize every combinatorial detail.

The workflow improved over time. Early in the process, AI was most useful for brainstorming structure and possible proof strategies. Later, I used it more selectively: mostly to tighten exposition, catch overstatements, and package the final work into a clean submission format. That made the final pass much more efficient.

Conclusion

AI was useful for organization, drafting, and sanity checks, but the final correctness of the assignment depended on explicit proofs, independent verification, and careful revision of any suggestions that were too loose.

Lean Formalization

Lean: NO.

Computational Check

The computational support mentioned here appears in `hw2/code/enumeration_check.py`, with outputs in `hw2/figures/`.

Repository Artifacts and Mapping

Supporting Repository Artifacts

- `hw2/writeup.tex`: source file for the write-up.
- `hw2/writeup.pdf`: compiled PDF draft.
- `hw2/lean/vc_dimension_bound.lean`: Lean target for Problem B1.
- `hw2/lean/quadratic_embedding.lean`: Lean target for Problem B2.
- `hw2/code/enumeration_check.py`: small-n verification for Problems A2 and B4.
- `hw2/figures/small_n_growth.csv`: table of computed values.

- `hw2/figures/small_n_growth.png`: small-n growth plot.

Write-Up to Repository Mapping

- Problem A2 → `hw2/code/enumeration_check.py`, `hw2/figures/small_n_growth.csv`
- Problem B1 → `hw2/lean/vc_dimension_bound.lean`
- Problem B2 → `hw2/lean/quadratic_embedding.lean`
- Problem B4 → `hw2/code/enumeration_check.py`, `hw2/figures/small_n_growth.png`
- Full submission draft → `hw2/writeup.tex` and `hw2/writeup.pdf`

Lean Targets

The most appropriate Lean targets in this assignment are:

- the general linear-representation VC-dimension upper bound from Problem B1,
- the quadratic embedding identity from Problem B2.

These are the cleanest and most reusable formal statements in the assignment.